# Lab 4: Redirection and Pipelines

```
Work directly in the terminal. No Bash script is required.
Goal: select records, extract fields, count lines, and distinguish normal output from errors.
120 minutes • individual account • personal status words
```

# Streams and operators

```
stdin = 0 • stdout = 1 • stderr = 2
> replaces stdout in a file
>> appends stdout
2> saves stderr separately
> file 2>&1 saves both streams
A pipe carries stdout only.
```

# Setup: 0–10 minutes

```
Run oslab start lab4 and enter its workspace using the supplied setup lines in the instruction.
oslab values lab4
cat data/events.csv
Replace STATUS and OTHER with your own words.
Never reset to fix one command.
```

# Guided: 10–25 minutes

```
printf 'first observation\n' > evidence/practice.txt
printf 'second observation\n' >> evidence/practice.txt
cat evidence/practice.txt
Which operator preserves the first line?
```

# Display and save

```
printf 'shown and saved\n' | tee evidence/tee-demo.txt
printf 'one more line\n' | tee -a evidence/tee-demo.txt
tee writes to the screen and a file.
tee -a preserves earlier file contents.
```

# Separate output from errors

```
cat data/not-present.csv > evidence/demo-out.txt 2> evidence/demo-error.txt
echo $?
Inspect both files. Expected: empty stdout, diagnostic on stderr.
Read exit status immediately; the next command changes it.
```

# Prediction: 25–35 minutes

```
Read the personal records. Then run oslab predict lab4, alone without AI.
Predict the first selected quantity, matching-record count and error stream.
A count is not the first quantity or a sum.
```

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# Core 1: select, extract, count • 35–55

```
grep '^STATUS,' data/events.csv | tee evidence/selected.txt
grep '^STATUS,' data/events.csv | cut -d, -f2 | tee evidence/fields.txt
grep '^STATUS,' data/events.csv | wc -l > evidence/normal.txt
The comma excludes STATUS-late.
Expected count: 3. Quantities are not added.
```

# Compare and record

```
diff -u data/expected.txt evidence/normal.txt
No output means identical contents.
Use nano to save actual commands in evidence/commands.txt, one per line.
This is a text record, not a script to execute.
oslab check lab4
```

## Core 2: changed inputs • 55–70

```
printf 'status,count\nOTHER,2\n' > data/no-match.csv
cp -- data/events.csv data/changed.csv
printf 'STATUS,7\nSTATUS-late,99\n' >> data/changed.csv
grep '^STATUS,' data/no-match.csv | wc -l > evidence/edge.txt
grep '^STATUS,' data/changed.csv | wc -l >> evidence/edge.txt
Expected: 0 then 4. The quantity 7 adds one record.
```

# Core 2: missing input

```
cat data/missing.csv > evidence/stdout.txt 2> evidence/stderr.txt
echo $? > evidence/status.txt
Expected: empty stdout, diagnostic naming missing.csv, nonzero status.
oslab check lab4
Four Core milestones check saved observations, not command authorship.
```

# Demonstration and optional work • 70–85

```
cat data/missing.csv | wc -l
echo $?
The last stage may succeed even when the input command fails.
Plus: inspect grep's no-match status (1).
Challenge: put stderr into the pipe and use tee to display/save both streams.
```

# Checkpoint: 85–100 minutes

```
Wait for release; close AI tools.
oslab checkpoint lab4
Answer questions about fresh records. Create the input with the supplied command.
Write your own exact-selection pipeline.
Save evidence/checkpoint-selected.txt and checkpoint.txt (record count).
oslab check lab4 should show 5/5.
```

# Debrief: 100–110 minutes

```
oslab debrief lab4
Why does wc -l count records here?
What does cut extract?
Why is the comma important in grep?
How did you prove a diagnostic used stderr?
Why can a pipeline report status 0 after an input error?
```

# Submit and grading: 110–120 minutes

```
Submit README.md, checkpoint.txt, and required evidence/*.txt from the instruction.
Include commands.txt and the empty stdout.txt. No report.sh.
Core results: 2 • commands/tests: 2 • prediction/debrief: 2 • checkpoint: 3 • submission: 1
Scripts belong in Lab 7. Advanced extensions require later prerequisites.
```

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